Co-Designing Inclusive Technologies

The goal of this assignment is to **co-design a computing artifact** that addresses a specific need or desire of a person from a minority group.

- **Deliverables:** 10-min presentation, research paper, and high-fidelity prototype
- **Duration**: 7 weeks
- **Deadline:**
  - Mid-term: Studio of week of October 5th
  - Final: Studio of week of October 19th

The design issues covered in this project are: (1) co-design, (2) inclusive design, (3) problem framing, (4) ideation, and (5) prototyping.

Inclusive Design is about including users that are often excluded from traditional design processes (e.g., people with disabilities, ethnic minorities, older adults, lower socio-economic status). Inclusive designers celebrate difference while aiming to cater for the full spectrum of human diversity. It is about empowering people by respecting and appreciating what makes them different, in terms of, e.g., age, gender, ethnicity, religion, sexual orientation, or (dis)ability.

This assignment aims to design a computing artifact alongside one co-designer from a minority group (external to the course). The project should address a specific need or desire of the co-designer; this poses several design challenges including problem framing, concept ideation & validation, co-design methods, inclusive design processes, prototyping, and testing.

Recruitment

You will start by identifying and recruiting one co-designer from a minority group. You should inform the co-designer what you will ask of them in the following weeks: **a total of ±3 hours split between 3 sessions**. These sessions should be conducted outside the classroom (i.e., autonomous work).

For the first time, this year, **faculty will provide access to two co-designers for four teams**. Two teams from the Friday shift at Alameda can request to work with **Associação Fermenta** (https://www.fermenta.org/pt/, making with older women), while two teams from the Tuesday shift at Taguspark can request to work with **ImpactSci** (https://impactsci.org/, neurodivergent advocacy and education). If you are enrolled in one of these shifts and want to work with these co-designers, send an email to hugo.nicolau@tecnico.ulisboa.pt (first come, first served). **All other teams will have to recruit a co-designer.**

Session 1: Explore the Problem Space

You are encouraged to use the Double Diamond design process (this will be covered in lectures and studios) to guide your work with the co-designer. In the first session, you will explore the problem space (i.e., first diamond) and create a **project declaration with your co-designer**. This document should (1) briefly describe your co-designer, (2) identify a specific need or opportunity, and (3) describe what the co-designer will benefit from this

project. **The co-designer needs to approve this document.** Both the design team and co-designer should sign the document. You can leverage one or a combination of research methods to uncover a need: ranging from interviews and directed storytelling to cognitive mapping and magic machines (see section Resources). Before this session, you should conduct secondary research (covered in Studio 1).

Session 2: Explore the Solution Space

In the second session, you should run a **co-design ideation workshop**, where you and the co-designer work together in defining one or more design concepts. This activity aims to "give voice" to your co-designer's needs, **making sure their priorities and values are reflected in the final design**. Your role is to support the co-designer's expression and provide all the resources needed to engage in a creative activity (**rather than provide your design to the "user"!**). These activities can include collages, storyboarding, experience prototyping, imagining future scenarios, and paper prototyping.

Session 3: Testing the Solution

After the co-design workshop, your team should engage in an internal iterative design process to create a high-fidelity prototype. **The high-fidelity prototype does not need to be functional**. It can be a video prototype, interactive Figma prototype, interactive object that is controlled via Wizard of Oz, etc. **The main goal of the prototype is to be "demoable" and allow you to collect feedback.** Once you have your high-fidelity prototype, you should conduct a **final session with the co-designer to collect their feedback.**

You are highly encouraged to invite your co-designer to the final in-class project presentation.

Schedule

This section suggests a schedule to tackle the design brief. **Feel free to adapt it to your project needs as long as you fulfill the project mid-term and final deadlines.**

Week 1: 07 – 11 Sep	Lectures: - Introduction - How do understand and define problems Studio: - Define teams - Assignment 1: Exploring the problem space	Autonomous Work: - **Recruit co-designer**
Week 2: 14 – 18 Sep	Lectures: - How to be creative - How to prototype Studio: - Assignment 2: Exploring the solution space	Autonomous Work: - **Finalise recruitment**
Week 3: 21 – 25 Sep	Lectures: - Inclusive design & Design brief - Invited research talks on Inclusive Design Studio:	Autonomous Work: - **Conduct 1st co-design session and agree on a project declaration**

	- Discuss design brief and co-designer - Prepare and pilot 1st co-design session - Peer critique	
Week 4: 28 Sep – 2 Oct	Lectures: - How to write a research paper - How do design user interfaces Studio: - Discuss project declaration - Prepare and pilot 2nd co-design session - Peer critique	Autonomous Work: - **Conduct 2nd co-design session** - Prepare mid-term deliverables
Week 5: 05 – 9 Oct	Lectures: - How to design the Internet of Things - Invited panel: IxD in Industry Studio: - Mid-term project presentation	Autonomous Work: - **Iterative prototyping**
Week 6: 12 – 16 Oct	Lectures: - Project support Studio: - Project support	Autonomous Work: - **Conduct 3rd co-design session** - **Reflect on findings**
Week 7: 19 – 23 Oct	Lecture: - Project support Studio: - Final project presentations** **** You are highly encouraged to invite your co-designer to the final project presentation ****	Autonomous Work: - **Prepare and submit final deliverables**

Deliverables

Mid-term deliverables

1. **5-minute presentation**: should include the problem framing (statement and argument), description of sessions 1 and 2 (i.e., used materials, activities, methods, and lessons learned).
2. **Project declaration document**: description of the co-designer; (user-centric) problem statement that identifies a specific need or opportunity; both the participant and the team need to approve the project declaration by signing it.
3. **First draft of research paper**: introduction, related work, methodology, secondary research, and early findings sections.

Presentations take place in studio 5. Both deliverables need to be submitted via Fenix by the end of week 5. Deliverables are not graded.

Final deliverables

1. **10-minute presentation**: should include the problem framing, description of the whole design process (focus on the methods!), the design concept and how it addresses the identified problem, demo of the prototype and how it evolved over time, lessons learned from 3rd session, and possible improvements.
2. **High-fidelity prototype**: *demoable* prototype (does not need to be a functional prototype)
3. **Research paper**: research paper describing your problem framing, process, prototype, and findings. Examples here.

Presentations take place in studio 7. For Thursday and Friday studios, presentations will take place on Tuesday and Wednesday lecture slot of the same week. Deliverables need to be submitted via Fenix by the end of week 7.

Grading Criteria

Only the final deliverables will be graded. The mid-term deliverables will be used by Faculty and classmates to provide constructive feedback and improve your work.

Most of the grading is related to the Design Process. Thus, you should carefully consider the design methods (see next section) used throughout the process to empower your co-designer. Leverage the studios to discuss your decisions with Faculty and peers. Feel free to combine and adapt multiple methods to your needs. ***It is crucial you demonstrate that the chosen methods enabled the co-designer had a "voice" throughout the process.***

Design Process (60%): Has the team identified a meaningful problem/opportunity? Did they include the co-designer in the problem framing process? Were the methods and materials appropriate for a co-design process? How did the team make sure the co-designer had an active voice throughout the process? Did it support the co-designer creative process? Were the lessons learned relevant for the project? How did the team engage in an iterative prototyping process?

Prototype (20%): Is the design innovative and useful? Does it effectively cover the identified need or opportunity? Does it reflect the "users' voice" from the co-design sessions?

Communication (10%): Clarity of communication and visual appeal of the deliverables;

Peer Grading (10%): Individual peer assessment. You can find the **assessment scale here**. **You should submit the peer grading form until the final deadline**.

The grade of the Design Process, Prototype, and Communication will be assessed through the Presentation and Research paper with the respective weight of 50%.

Resources

- What is co-design and why is it useful?
- Explore dozens of design methods and tools
- Examples of co-design methods
- Design. Think. Make. Break. Repeat. [WEB][FNAC][Amazon] *Very good book that concisely presents multiple design methods*
- The Art of Co-Design: Solving Problems through Creative Collaboration. [Amazon]

Other Design Methods:

- Directed storytelling, Cognitive mapping, Card sorting, Design Workshop